

Intro

This document is to look at how our refactoring examples affected the CK metrics.

Metrics Before

22 types	WMC	DIT	NOC	CBO	RFC	Lack of Cohesion Of Methods (LCOM)	Methods
22 types matched							
▲ FitsMetrics (22 types)							
▲ {} (19 types)							
AstTool	0	1	0	11	0	0	23
DataAnalysis	13	N/A	0	39	22	1.0	4
FitsReader	55	1	0	27	126	1.0	61

So starting with weighted methods per class, these are all adaptors so our goal is to reduce these to below 40, FITS reader is the only one out of spec.

Depth of inheritance tree, these inherent from System.Object so DIT is 1 for the ASTTool and FITS Reader, it is N/A for DataAnalysis because it is a static class, inheritance doesn't apply to those.

Number of Children, zero for all, no class extends these.

Coupling Between Object Classes, these are all adaptors so < 25 is the goal, it's tricky to reduce this number because these plugins whole purpose is to have their methods called from other classes.

Response for a class, FltsReader is the only one here that is out of spec, will be reduced by separating methods into individual specialised classes based on their function.

Lack of Cohesion Of Methods, this is 1.0 for the DataAnalysis and FitsReader, the worst possible score, this really signifies that the FitsReader needed to be recast into individual classes that handle their own specific tasks.

Metrics After

33 types	WMC	DIT	NOC	CBO	RFC	Lack of Cohesion Of Methods (LCOM)	Methods
33 types matched							
▲ iDeVIE.PluginInterface (33 types)							
▲ {} (25 types)							
DefaultExecutionOrderAttribute	1	2	0	6	0	0	2
FitsFile	2	N/A	0	7	2	0	16
FitsHeader	5	N/A	0	13	20	0	15
FitsImage	0	N/A	0	7	0	0	15
FitsTable	7	N/A	0	12	28	0	5
FitsMask	41	N/A	0	19	82	0	9
AstTool	0	1	0	9	0	0	23
DataAnalysis	12	N/A	0	38	25	0	5

Brief Specs

7.1 Chidamber and Kemerer (CK) suite (Mandatory)

CK metric	Meaning	Acceptable range
WMC	Weighted Methods per Class.	≤ 20 domain; ≤ 40 adapters.
DIT	Depth of Inheritance Tree.	≤ 4 .
NOC	Number of Children.	≤ 5 .
CBO	Coupling Between Object classes.	≤ 14 domain; ≤ 25 orchestrators. Cycles forbidden.
RFC	Response For a Class.	≤ 50 .
LCOM	Lack of Cohesion in Methods (Henderson-Sellers variant).	≤ 0.5 .